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This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

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## Device Specifications

Device Type: Camera  
Manufacturer: Array Telepresence  
Firmware Version: N/A  
Model(s): Equal-i 2S

## Tested on the Following Software and Firmware Versions

| IP Link Pro Control Processor Firmware | Global Scriptor Version |
|----------------------------------------|-------------------------|
| 2.06.0002-b002                         | 1.4.2                   |

## Version History

| Module Version | Date     | Notes           |
|----------------|----------|-----------------|
| 1_0_0_0        | 7/2/2018 | Initial Version |

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## Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.  
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.  
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword must be set accordingly. devicePassword default value is set to 'ATEquali'.  
Example: `InterfaceName.devicePassword = 'extron'`

## Supported Classes and Examples

|                      |
|----------------------|
| <b>EthernetClass</b> |
|----------------------|

|                                                                                                    |
|----------------------------------------------------------------------------------------------------|
| <code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 2000, Model='Equal-i 2S')</code> |
|----------------------------------------------------------------------------------------------------|

## Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

|                                                                                                 |                          |                          |                 |
|-------------------------------------------------------------------------------------------------|--------------------------|--------------------------|-----------------|
| Command                                                                                         | Value                    | Value                    | Value           |
| <b>BackEndOperatingMode</b>                                                                     | 'Passthrough'            | 'Immersive Mode Receive' | 'Immersive PTZ' |
| # BackEndOperatingMode example<br>InterfaceName.Set('BackEndOperatingMode', 'Passthrough')      |                          |                          |                 |
| Command                                                                                         | Value                    |                          |                 |
| <b>BalanceBoxPosition</b>                                                                       | 1 to 100 in steps of 1   |                          |                 |
| # BalanceBoxPosition example<br>InterfaceName.Set('BalanceBoxPosition', 100)                    |                          |                          |                 |
| Command                                                                                         | Value                    |                          |                 |
| <b>ColorGain</b>                                                                                | 0 to 65535 in steps of 1 |                          |                 |
| Qualifier Key                                                                                   | Qualifier Value          | Qualifier Value          |                 |
| 'Camera'                                                                                        | 'Left'                   | 'Right'                  |                 |
| # ColorGain example<br>InterfaceName.Set('ColorGain', 65535, {'Camera': 'Left'})                |                          |                          |                 |
| Command                                                                                         | Value                    | Value                    | Value           |
| <b>FrontEndOperatingMode</b>                                                                    | 'Immersive Mode'         | 'Video+Content'          | 'Stack'         |
|                                                                                                 | 'Immersive Everywhere'   |                          |                 |
| # FrontEndOperatingMode example<br>InterfaceName.Set('FrontEndOperatingMode', 'Immersive Mode') |                          |                          |                 |
| Command                                                                                         | Value                    |                          |                 |
| <b>Gamma</b>                                                                                    | '1' – '6'                |                          |                 |
| # Gamma example<br>InterfaceName.Set('Gamma', '1')                                              |                          |                          |                 |
| Command                                                                                         | Value                    |                          |                 |
| <b>IntegrationTime</b>                                                                          | 1 to 1640 in steps of 1  |                          |                 |
| Qualifier Key                                                                                   | Qualifier Value          | Qualifier Value          |                 |
| 'Camera'                                                                                        | 'Left'                   | 'Right'                  |                 |
| # IntegrationTime example<br>InterfaceName.Set('IntegrationTime', 1640, {'Camera': 'Left'})     |                          |                          |                 |
| Command                                                                                         | Value                    |                          |                 |
| <b>Pan</b>                                                                                      | 1 to 1000 in steps of 1  |                          |                 |
| Qualifier Key                                                                                   | Qualifier Value          | Qualifier Value          |                 |
| 'Camera'                                                                                        | 'Left'                   | 'Right'                  |                 |
| Qualifier Key                                                                                   | Qualifier Value          | Qualifier Value          |                 |

|                                                                                                |                           |                            |
|------------------------------------------------------------------------------------------------|---------------------------|----------------------------|
| <b>'Direction'</b>                                                                             | <b>'Horizontal'</b>       | <b>'Vertical'</b>          |
| # Pan example<br>InterfaceName.Set('Pan', 1000, {'Camera': 'Left', 'Direction': 'Horizontal'}) |                           |                            |
| <b>Command</b><br><b>Power</b>                                                                 | Value<br>'On'             | Value<br>'Off'             |
| # Power example<br>InterfaceName.Set('Power', 'On')                                            |                           |                            |
| <b>Command</b><br><b>SensorGain</b>                                                            | Value<br>'x1'<br>'x8'     | Value<br>'x2'<br>'x4'      |
| <b>Qualifier Key</b><br><b>'Camera'</b>                                                        | Qualifier Value<br>'Left' | Qualifier Value<br>'Right' |
| # SensorGain example<br>InterfaceName.Set('SensorGain', 'x1', {'Camera': 'Left'})              |                           |                            |
| <b>Command</b><br><b>WhiteBalanceBoxes</b>                                                     | Value<br>'On'             | Value<br>'Off'             |
| # WhiteBalanceBoxes example<br>InterfaceName.Set('WhiteBalanceBoxes', 'On')                    |                           |                            |

## Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

|                                                                                                                                                                                                                             |                          |                          |                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|-----------------|
| Command                                                                                                                                                                                                                     | Value                    | Value                    | Value           |
| BackEndOperatingMode                                                                                                                                                                                                        | 'Passthrough'            | 'Immersive Mode Receive' | 'Immersive PTZ' |
| # BackEndOperatingMode examples<br>InterfaceName.Update('BackEndOperatingMode')<br>Value = InterfaceName.ReadStatus('BackEndOperatingMode')<br>InterfaceName.SubscribeStatus('BackEndOperatingMode', None, FeedbackHandler) |                          |                          |                 |
| Command                                                                                                                                                                                                                     | Value                    |                          |                 |
| BalanceBoxPosition                                                                                                                                                                                                          | 1 to 100 in steps of 1   |                          |                 |
| # BalanceBoxPosition examples<br>InterfaceName.Update('BalanceBoxPosition')<br>Value = InterfaceName.ReadStatus('BalanceBoxPosition')<br>InterfaceName.SubscribeStatus('BalanceBoxPosition', None, FeedbackHandler)         |                          |                          |                 |
| Command                                                                                                                                                                                                                     | Value                    |                          |                 |
| ColorGain                                                                                                                                                                                                                   | 0 to 65535 in steps of 1 |                          |                 |
| Qualifier Key                                                                                                                                                                                                               | Qualifier Value          | Qualifier Value          |                 |
| 'Camera'                                                                                                                                                                                                                    | 'Left'                   | 'Right'                  |                 |
| # ColorGain examples<br>InterfaceName.Update('ColorGain', {'Camera': 'Left'})<br>Value = InterfaceName.ReadStatus('ColorGain', {'Camera': 'Left'})<br>InterfaceName.SubscribeStatus('ColorGain', None, FeedbackHandler)     |                          |                          |                 |
| Command                                                                                                                                                                                                                     | Value                    | Value                    |                 |
| ConnectionStatus                                                                                                                                                                                                            | 'Connected'              | 'Disconnected'           |                 |
| # ConnectionStatus examples<br>Value = InterfaceName.ReadStatus('ConnectionStatus')<br>InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)                                                             |                          |                          |                 |
| Command                                                                                                                                                                                                                     | Value                    | Value                    | Value           |
| FrontEndOperatingMode                                                                                                                                                                                                       | 'Immersive Mode'         | 'Video+Content'          | 'Stack'         |

|                                                                                                                                                                                                                                                       |  |                                  |                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------|----------------------------------------|
|                                                                                                                                                                                                                                                       |  | 'Immersive Everywhere'           |                                        |
| # FrontEndOperatingMode examples<br>InterfaceName.Update('FrontEndOperatingMode')<br>Value = InterfaceName.ReadStatus('FrontEndOperatingMode')<br>InterfaceName.SubscribeStatus('FrontEndOperatingMode', None, FeedbackHandler)                       |  |                                  |                                        |
| Command<br>Gamma                                                                                                                                                                                                                                      |  | Value<br>'1' – '6'               |                                        |
| # Gamma examples<br>InterfaceName.Update('Gamma')<br>Value = InterfaceName.ReadStatus('Gamma')<br>InterfaceName.SubscribeStatus('Gamma', None, FeedbackHandler)                                                                                       |  |                                  |                                        |
| Command<br>IntegrationTime                                                                                                                                                                                                                            |  | Value<br>1 to 1640 in steps of 1 |                                        |
| Qualifier Key<br>'Camera'                                                                                                                                                                                                                             |  | Qualifier Value<br>'Left'        | Qualifier Value<br>'Right'             |
| # IntegrationTime examples<br>InterfaceName.Update('IntegrationTime', {'Camera': 'Left'})<br>Value = InterfaceName.ReadStatus('IntegrationTime', {'Camera': 'Left'})<br>InterfaceName.SubscribeStatus('IntegrationTime', None, FeedbackHandler)       |  |                                  |                                        |
| Command<br>Pan                                                                                                                                                                                                                                        |  | Value<br>1 to 1000 in steps of 1 |                                        |
| Qualifier Key<br>'Camera'                                                                                                                                                                                                                             |  | Qualifier Value<br>'Left'        | Qualifier Value<br>'Right'             |
| Qualifier Key<br>'Direction'                                                                                                                                                                                                                          |  | Qualifier Value<br>'Horizontal'  | Qualifier Value<br>'Vertical'          |
| # Pan examples<br>InterfaceName.Update('Pan', {'Camera': 'Left', 'Direction': 'Horizontal'})<br>Value = InterfaceName.ReadStatus('Pan', {'Camera': 'Left', 'Direction': 'Horizontal'})<br>InterfaceName.SubscribeStatus('Pan', None, FeedbackHandler) |  |                                  |                                        |
| Command<br>Power                                                                                                                                                                                                                                      |  | Value<br>'On'                    | Value<br>'Off'                         |
| # Power examples<br>InterfaceName.Update('Power')<br>Value = InterfaceName.ReadStatus('Power')<br>InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)                                                                                       |  |                                  |                                        |
| Command<br>SensorGain                                                                                                                                                                                                                                 |  | Value<br>'x1'<br><br>'x8'        | Value<br>'x2'<br><br><br>Value<br>'x4' |
| Qualifier Key<br>'Camera'                                                                                                                                                                                                                             |  | Qualifier Value<br>'Left'        | Qualifier Value<br>'Right'             |
| # SensorGain examples<br>InterfaceName.Update('SensorGain', {'Camera': 'Left'})<br>Value = InterfaceName.ReadStatus('SensorGain', {'Camera': 'Left'})<br>InterfaceName.SubscribeStatus('SensorGain', None, FeedbackHandler)                           |  |                                  |                                        |

| Command                                                                                                                                                                                                           | Value | Value |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| <b>WhiteBalanceBoxes</b>                                                                                                                                                                                          | 'On'  | 'Off' |
| <pre># WhiteBalanceBoxes examples InterfaceName.Update('WhiteBalanceBoxes') Value = InterfaceName.ReadStatus('WhiteBalanceBoxes') InterfaceName.SubscribeStatus('WhiteBalanceBoxes', None, FeedbackHandler)</pre> |       |       |

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## Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

|                                     |              |
|-------------------------------------|--------------|
| <b>Port Type:</b>                   | Ethernet     |
| <b>Default Port:</b>                | 2000         |
| <b>Logon Credentials Supported:</b> | Yes          |
| <b>Default Password:</b>            | ATEquali     |
| <b>Multi-Connection</b>             | Undetermined |
| <b>Capabilities:</b>                |              |
| <b>Port Changeability:</b>          | No           |

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## Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

## Notes for the Device



## Appendix A. Set Commands

|                                                |                         |
|------------------------------------------------|-------------------------|
| Back End Operating Mode Immersive Mode Receive | :OPMODEBE=2\ \x0D       |
| Back End Operating Mode Immersive PTZ          | :OPMODEBE=3\ \x0D       |
| Back End Operating Mode Passthrough            | :OPMODEBE=1\ \x0D       |
| Balance Box Position 1                         | :BALBOXPOS=1\ \x0D      |
| Balance Box Position 100                       | :BALBOXPOS=100\ \x0D    |
| Color Gain 0 Camera Left                       | :COLGAIN=0, L\ \x0D     |
| Color Gain 65535 Camera Left                   | :COLGAIN=65535, L\ \x0D |
| Color Gain 0 Camera Right                      | :COLGAIN=0, R\ \x0D     |
| Color Gain 65535 Camera Right                  | :COLGAIN=65535, R\ \x0D |
| Front End Operating Mode Immersive Everywhere  | :OPMODEFE=4\ \x0D       |
| Front End Operating Mode Immersive Mode        | :OPMODEFE=1\ \x0D       |
| Front End Operating Mode Stack                 | :OPMODEFE=3\ \x0D       |
| Front End Operating Mode Video+Content         | :OPMODEFE=2\ \x0D       |
| Gamma 1                                        | :GAMMA=1\ \x0D          |
| Gamma 6                                        | :GAMMA=6\ \x0D          |
| Integration Time 1 Camera Left                 | :INTTIME=1, L\ \x0D     |
| Integration Time 1640 Camera Left              | :INTTIME=1640, L\ \x0D  |
| Integration Time 1 Camera Right                | :INTTIME=1, R\ \x0D     |
| Integration Time 1640 Camera Right             | :INTTIME=1640, R\ \x0D  |
| Pan 1 Camera Left Direction Horizontal         | :HORPAN=1, L\ \x0D      |
| Pan 1000 Camera Left Direction Horizontal      | :HORPAN=1000, L\ \x0D   |
| Pan 1 Camera Left Direction Vertical           | :VERTPAN=1, L\ \x0D     |
| Pan 1000 Camera Left Direction Vertical        | :VERTPAN=1000, L\ \x0D  |
| Pan 1 Camera Right Direction Horizontal        | :HORPAN=1, R\ \x0D      |
| Pan 1000 Camera Right Direction Horizontal     | :HORPAN=1000, R\ \x0D   |
| Pan 1 Camera Right Direction Vertical          | :VERTPAN=1, R\ \x0D     |
| Pan 1000 Camera Right Direction Vertical       | :VERTPAN=1000, R\ \x0D  |
| Power Off                                      | :SLEEP=1\ \x0D          |
| Power On                                       | :SLEEP=0\ \x0D          |
| Sensor Gain x1 Camera Left                     | :SENSGAIN=1, L\ \x0D    |
| Sensor Gain x2 Camera Left                     | :SENSGAIN=2, L\ \x0D    |
| Sensor Gain x4 Camera Left                     | :SENSGAIN=3, L\ \x0D    |
| Sensor Gain x8 Camera Left                     | :SENSGAIN=4, L\ \x0D    |
| Sensor Gain x1 Camera Right                    | :SENSGAIN=1, R\ \x0D    |
| Sensor Gain x2 Camera Right                    | :SENSGAIN=2, R\ \x0D    |
| Sensor Gain x4 Camera Right                    | :SENSGAIN=3, R\ \x0D    |

## Global Scripter Module Communication Sheet

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|                                    |                   |
|------------------------------------|-------------------|
| <b>Sensor Gain x8 Camera Right</b> | :SENSGAIN=4,R\x0D |
| <b>White Balance Boxes Off</b>     | :BALBOXES=0\x0D   |
| <b>White Balance Boxes On</b>      | :BALBOXES=1\x0D   |

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**Appendix B. Update Commands**

|                                              |                 |
|----------------------------------------------|-----------------|
| <b>Back End Operating Mode</b>               | :OPMODEBE?\x0D  |
| <b>Balance Box Position</b>                  | :BALBOXPOS?\x0D |
| <b>Color Gain Camera Left</b>                | :COLGAIN?L\x0D  |
| <b>Color Gain Camera Right</b>               | :COLGAIN?R\x0D  |
| <b>Front End Operating Mode</b>              | :OPMODEFE?\x0D  |
| <b>Gamma</b>                                 | :GAMMA?\x0D     |
| <b>Integration Time Camera Left</b>          | :INTTIME?L\x0D  |
| <b>Integration Time Camera Right</b>         | :INTTIME?R\x0D  |
| <b>Pan Camera Left Direction Horizontal</b>  | :HORPAN?L\x0D   |
| <b>Pan Camera Left Direction Vertical</b>    | :VERTPAN?L\x0D  |
| <b>Pan Camera Right Direction Horizontal</b> | :HORPAN?R\x0D   |
| <b>Pan Camera Right Direction Vertical</b>   | :VERTPAN?R\x0D  |
| <b>Power</b>                                 | :SLEEP?\x0D     |
| <b>Sensor Gain Camera Left</b>               | :SENSGAIN?L\x0D |
| <b>White Balance Boxes</b>                   | :BALBOXES?\x0D  |